

FleetSpark — API Specification

Version: 1.0

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Base URL: `https://api.fleetspark.syncspherellc.com`

Protocol: REST + WebSocket

Auth: JWT (Bearer token) + API Key (telemetry)

1. Authentication

POST /auth/register

Register a new organization + admin user.

Request:

```
{
  "orgName": "Ugandan EV manufacturers",
  "orgSlug": "kiira-motors",
  "userName": "Fleet Admin",
  "email": "admin@kiiramotors.com",
  "password": "securePassword123"
}
```

Response (201):

```
{
  "user": {
    "id": "uuid",
    "email": "admin@kiiramotors.com",
    "name": "Fleet Admin",
    "role": "ADMIN",
    "orgId": "uuid"
  },
  "tokens": {
    "accessToken": "eyJ...",
    "refreshToken": "eyJ...",
    "expiresIn": 900
  }
}
```

POST /auth/login

Request:

```
{
  "email": "admin@kiiramotors.com",
  "password": "securePassword123"
}
```

Response (200):

```
{
  "user": { "id": "uuid", "email": "...", "name": "...", "role": "ADMIN", "orgId": "uuid" },
  "tokens": { "accessToken": "eyJ...", "refreshToken": "eyJ...", "expiresIn": 900 }
}
```

POST /auth/refresh

Request:

```
{
  "refreshToken": "eyJ..."
}
```

Response (200):

```
{
  "accessToken": "eyJ...",
  "refreshToken": "eyJ...",
  "expiresIn": 900
}
```

POST /auth/logout

Headers: Authorization: Bearer <accessToken>

Response (204): No content

2. Vehicles

All endpoints require: Authorization: Bearer <accessToken>

GET /api/v1/vehicles

List all vehicles for the organization.

Query Parameters:

Param	Type	Default	Description
page	int	1	Page number
limit	int	25	Items per page (max 100)
status	string	—	Filter by status
type	string	—	Filter by type
search	string	—	Search by name or plate

Response (200):

```
{
  "data": [
    {
      "id": "uuid",
      "name": "ElectricBus-001",
      "plateNumber": "UAX 123A",
      "model": "electric buses",
      "type": "BUS",
      "status": "IN_TRANSIT",
      "batteryLevel": 72,
      "soh": 95,
      "estimatedRangeKm": 216,
      "lastLocation": { "lat": 0.3136, "lng": 32.5811 },
      "lastUpdate": "2026-06-11T14:30:00Z",
      "driver": { "id": "uuid", "name": "John Kato" }
    }
  ],
  "meta": {
    "total": 24,
    "page": 1,
    "limit": 25,
    "totalPages": 1
  }
}
```

POST /api/v1/vehicles

Create a new vehicle. **Roles:** ADMIN, FLEET_MANAGER

Request:

```
{
  "name": "ElectricBus-001",
  "plateNumber": "UAX 147A",
  "vin": "KIIIRA202500147",
  "model": "electric buses",
  "year": 2025,
  "type": "BUS",
  "batteryCapacity": 250,
  "maxRangeKm": 300,
  "odometerKm": 0,
  "depot": "Kampala Central",
  "deviceId": "teltonika-abc123"
}
```

Response (201): Vehicle object

GET /api/v1/vehicles/:id

Response (200): Full vehicle object with latest telemetry

```
{
  "id": "uuid",
  "name": "ElectricBus-001",
  "plateNumber": "UAX 123A",
  "vin": "KIIRA202500123",
  "model": "electric buses",
  "year": 2025,
  "type": "BUS",
  "status": "IN_TRANSIT",
  "batteryCapacity": 250,
  "maxRangeKm": 300,
  "odometerKm": 15234.5,
  "depot": "Kampala Central",
  "deviceId": "teltonika-abc123",
  "driver": { "id": "uuid", "name": "John Kato", "phone": "+256772111111" },
  "currentTelemetry": {
    "lat": 0.3136,
    "lng": 32.5811,
    "speedKmh": 45,
    "batteryLevel": 72,
    "soh": 95,
    "estimatedRangeKm": 216,
    "charging": false,
    "temperature": 32,
    "lastUpdate": "2026-06-11T14:30:00Z"
  },
  "createdAt": "2025-01-15T10:00:00Z",
  "updatedAt": "2026-06-11T14:30:00Z"
}
```

PUT /api/v1/vehicles/:id

Update vehicle. **Roles:** ADMIN, FLEET_MANAGER
Request: Partial vehicle object (only fields to update)
Response (200): Updated vehicle object

DELETE /api/v1/vehicles/:id

Soft-delete vehicle. **Roles:** ADMIN only
Response (204): No content

GET /api/v1/vehicles/:id/location-history

Get location history for a vehicle.

Query Parameters:

Param	Type	Default	Description
from	ISO date	24h ago	Start time
to	ISO date	now	End time
interval	string	1m	Downsampling: 1m, 5m, 1h

Response (200):

```
{
  "data": [
    { "time": "2026-06-11T14:30:00Z", "lat": 0.3136, "lng": 32.5811, "speedKmh": 45 },
    { "time": "2026-06-11T14:29:00Z", "lat": 0.3130, "lng": 32.5805, "speedKmh": 42 }
  ],
  "meta": { "total": 1440, "downsampled": 288 }
}
```

3. Telemetry

POST /api/v1/telemetry

Auth: API Key header X-API-Key: <telemetry_api_key>

No JWT required — this endpoint is for GPS devices.

Rate Limit: 1 request per 30 seconds per vehicle

Request:

```
{
  "deviceId": "teltonika-abc123",
  "timestamp": "2026-06-11T14:30:00Z",
  "gps": {
    "lat": 0.3136,
    "lng": 32.5811,
    "speedKmh": 45,
    "heading": 180,
    "altitudeM": 1200
  },
  "ignition": true,
  "odometerKm": 15234.5,
  "battery": {
    "levelPct": 72,
    "sohPct": 95,
    "voltageV": 380,
    "temperatureC": 32,
    "charging": false,
    "currentA": -15.2
  }
}
```

Response (202):

```
{
  "received": true,
  "vehicleId": "uuid"
}
```

Errors:

- 401 — Invalid API key
- 404 — Device not found / not assigned to vehicle
- 429 — Rate limit exceeded (throttle)
- 422 — Validation error

4. Drivers

GET /api/v1/drivers

Query: page, limit, search, isActive

Response: Paginated driver list

POST /api/v1/drivers

Roles: ADMIN, FLEET_MANAGER

Request:

```
{
  "name": "Peter Mugisha",
  "phone": "+256772444444",
  "email": "peter@kiiramotors.com",
  "licenseNumber": "DL004",
  "licenseExpiry": "2027-12-31"
}
```

GET /api/v1/drivers/:id

PUT /api/v1/drivers/:id

DELETE /api/v1/drivers/:id (soft-delete)

POST /api/v1/drivers/:id/assign

Assign driver to vehicle.

```
{ "vehicleId": "uuid" }
```

POST /api/v1/drivers/:id/unassign

Unassign driver from vehicle.

Response (200): Updated driver object

5. Alerts

GET /api/v1/alerts

Query Parameters:

Param	Type	Default	Description
page	int	1	
limit	int	25	
type	string	—	Filter by alert type
severity	string	—	Filter by severity
vehicleId	string	—	Filter by vehicle

acknowledged	boolean	—	Filter by status
from	ISO date	7d ago	
to	ISO date	now	

Response (200):

```
{
  "data": [
    {
      "id": "uuid",
      "type": "LOW_BATTERY",
      "severity": "WARNING",
      "message": "ElectricBus-001 battery level at 18%",
      "vehicleId": "uuid",
      "vehicleName": "ElectricBus-001",
      "data": { "batteryLevelPct": 18 },
      "isAcknowledged": false,
      "createdAt": "2026-06-11T12:00:00Z"
    }
  ],
  "meta": { "total": 5, "page": 1, "limit": 25 }
}
```

POST /api/v1/alerts/:id/acknowledge

Roles: ADMIN, FLEET_MANAGER

Response (200): Updated alert object

GET /api/v1/alerts/settings

Get alert settings for the organization.

PUT /api/v1/alerts/settings/:type

Update alert threshold for a type.

```
{
  "threshold": { "level_pct": 25 },
  "enabled": true,
  "notifyEmail": true,
  "notifySms": false,
  "notifyInApp": true
}
```

6. Maintenance

GET /api/v1/maintenance

Query: page, limit, vehicleId, type, status, from, to

POST /api/v1/maintenance

Request:

```
{
  "vehicleId": "uuid",
  "type": "ROUTINE",
  "description": "Monthly brake inspection",
  "scheduledDate": "2026-06-15T09:00:00Z",
  "odometerKm": 16000,
  "notes": "Check front brake pads"
}
```

PUT /api/v1/maintenance/:id

Update maintenance record.

POST /api/v1/maintenance/:id/complete

Mark maintenance as completed.

```
{
  "completedDate": "2026-06-15T11:30:00Z",
  "costUgx": 150000,
  "notes": "Brake pads replaced"
}
```

GET /api/v1/maintenance/upcoming

Get upcoming maintenance (next 7 days).

Query: vehicleId, days (default 7)

7. Trips

GET /api/v1/trips

Query: page, limit, vehicleId, driverId, from, to

Response (200):


```
{
  "data": [
    {
      "id": "uuid",
      "vehicleId": "uuid",
      "vehicleName": "ElectricBus-001",
      "driverName": "John Kato",
      "startTime": "2026-06-11T08:00:00Z",
      "endTime": "2026-06-11T09:30:00Z",
      "startLat": 0.3136,
      "startLng": 32.5811,
      "endLat": 0.3456,
      "endLng": 32.6123,
      "distanceKm": 28.5,
      "avgSpeedKmh": 35,
      "maxSpeedKmh": 65,
      "durationMin": 90
    }
  ],
  "meta": { "total": 156 }
}
```

GET /api/v1/trips/:id

Full trip detail with route points.

8. Reports

GET /api/v1/reports/fleet-summary

Query: from, to (default: last 30 days)

Response (200):

```
{
  "period": { "from": "2026-05-11", "to": "2026-06-11" },
  "fleet": {
    "totalVehicles": 24,
    "totalDistanceKm": 45230,
    "totalTrials": 1850,
    "avgUtilizationPct": 78,
    "totalEnergyKwh": 8500,
    "totalChargingSessions": 340
  },
  "savings": {
    "fuelLitersSaved": 12500,
    "fuelCostSavedUgx": 68750000,
    "co2KgSaved": 33500,
    "maintenanceCostSavedUgx": 5000000
  },
  "vehicles": [
    {
      "id": "uuid",
      "name": "ElectricBus-001",
      "distanceKm": 2100,
      "trips": 85,
      "utilizationPct": 82,
      "energyKwh": 380,
      "fuelSavedUgx": 3100000
    }
  ]
}
```

GET /api/v1/reports/export

Query: type (fleet-summary | trips | maintenance), format (csv | pdf), from, to

Response: File download (CSV or PDF)

9. Users (Admin only)

GET /api/v1/users

List users in the organization.

POST /api/v1/users/invite

Invite a new user.

```
{
  "email": "manager@kiiramotors.com",
  "name": "Jane Manager",
  "role": "FLEET_MANAGER"
}
```

PUT /api/v1/users/:id/role

Change user role.

```
{ "role": "FLEET_MANAGER" }
```

DELETE /api/v1/users/:id

Deactivate user.

10. Settings

GET /api/v1/settings/organization

PUT /api/v1/settings/organization

```
{
  "name": "Ugandan EV manufacturers Corporation",
  "address": "Jinja Industrial Park, Jinja, Uganda",
  "phone": "+256414000000",
  "settings": {
    "defaultDieselPrice": 5500,
    "defaultEmissionFactor": 2.68,
    "timezone": "Africa/Kampala"
  }
}
```

POST /api/v1/settings/telemetry-key

Generate a new telemetry API key for GPS devices.

```
{ "name": "Teltonika Fleet A" }
```

Response:

```
{
  "id": "uuid",
  "name": "Teltonika Fleet A",
  "apiKey": "fs_tel_abc123...", // Show only once
  "createdAt": "2026-06-11T14:30:00Z"
}
```

11. WebSocket Events

Connection: `wss://api.fleetspark.syncsphere11c.com/ws`

Auth: Pass JWT token as query param: `?token=<accessToken>`

Client → Server Events

Event	Payload	Description
subscribe:vehicle	{ vehicleId: "uuid" }	Subscribe to real-time updates for a vehicle
subscribe:fleet	—	Subscribe to all vehicle updates in org
unsubscribe:vehicle	{ vehicleId: "uuid" }	Unsubscribe from vehicle
ping	—	Keep-alive

Server → Client Events

Event	Payload	Description
vehicle:location	{ vehicleId, lat, lng, speedKmh, batteryLevel, timestamp }	Real-time location update
vehicle:status	{ vehicleId, status, timestamp }	Status change (in_transit, charging, etc.)
vehicle:battery	{ vehicleId, levelPct, sohPct, charging, estimatedRangeKm }	Battery update
alert:new	{ alert: { id, type, severity, message, vehicleId, ... } }	New alert
alert:acknowledged	{ alertId, acknowledgedBy, acknowledgedAt }	Alert acknowledged
trip:started	{ tripId, vehicleId, driverId, startTime }	Trip started
trip:ended	{ tripId, distanceKm, durationMin }	Trip ended
pong	—	Keep-alive response

12. Error Format

All errors follow this structure:

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Request validation failed",
    "details": [
      { "field": "email", "message": "Invalid email format" }
    ]
  }
}
```

Error Codes

Code	HTTP Status	Description
UNAUTHORIZED	401	Missing or invalid token
FORBIDDEN	403	Insufficient permissions
NOT_FOUND	404	Resource not found
VALIDATION_ERROR	422	Invalid request data
RATE_LIMITED	429	Too many requests
INTERNAL_ERROR	500	Server error

13. Rate Limits

Endpoint Type	Limit	Window
Auth	10 requests	per minute per IP
API (authenticated)	1000 requests	per minute per user
Telemetry	1 request	per 30 seconds per vehicle
WebSocket	1 connection	per user

14. Pagination

All list endpoints use cursor-style pagination:

Request: `?page=1&limit=25`

Response:

```
{
  "data": [...],
  "meta": {
    "total": 100,
    "page": 1,
    "limit": 25,
    "totalPages": 4
  }
}
```